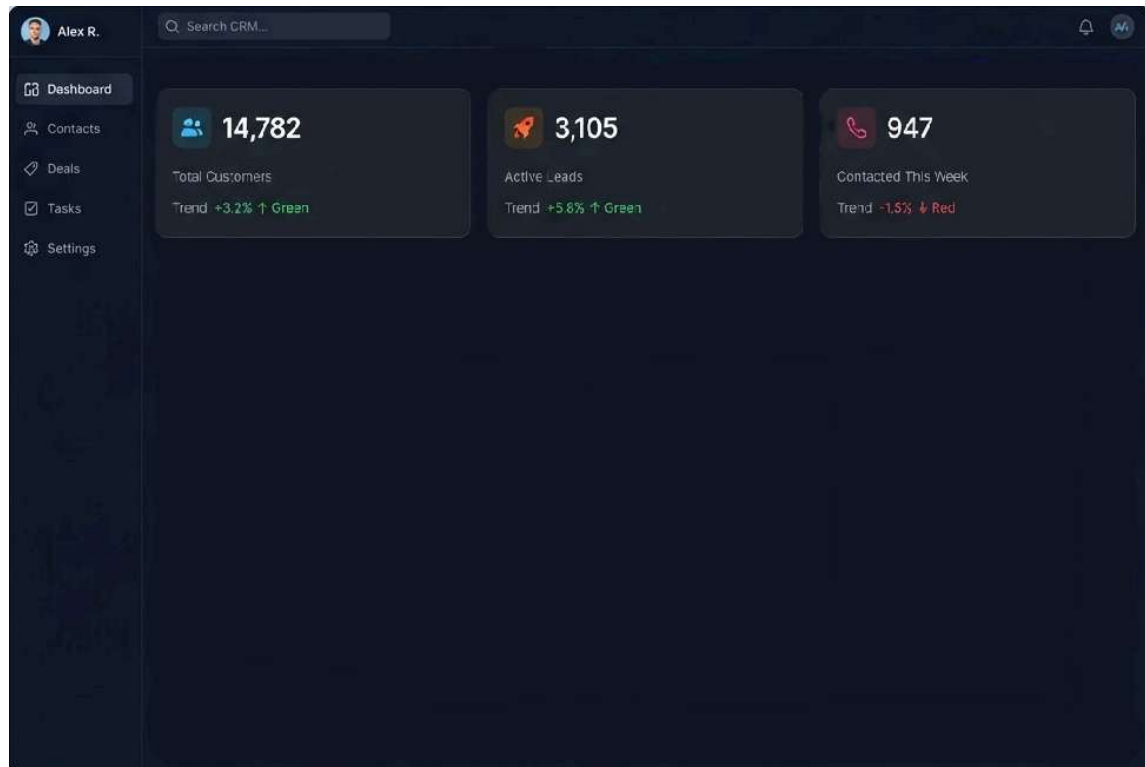


Advanced CRM Dashboard

Task Overview

Build a professional customer management dashboard with advanced filtering, powerful data management, and real-time search capabilities. This task evaluates proficiency with modern React/Next.js tooling, data fetching patterns, and component architecture.



Required Tech Stack (Must Use)

- **Next.js** (App Router)
- **TypeScript**
- **Tailwind CSS**
- **shadcn/ui** components
- **TanStack Query (React Query)** for data fetching & caching
- **Drag and drop** library (react-beautiful-dnd, dnd-kit, or similar) — for reorderable customer list or other interactive feature

Core Requirements

1. Customer List View with Search

- Display customers in a clean table or card layout
- Show: Name, Email, Phone, Company, Status (Active/Inactive), Last Contact Date
- Real-time search by name, email, or company
- Responsive design (works on mobile)
- Sorting by column (name, email, last contact date)
- Pagination (10, 25, 50 per page)

Customers

Search customers...

Status: All

Company: All

Add Customer

Name	Email	Phone	Company	Status	Last Contact	
 Alice Green	alicegreen@gmail.com	874-748-8877	Acme Corp	Active	Nov 12, 2023 - Nov 3, 2023	
 Bob Ross	bobross.coh@email.com	874-855-2469	Globex	Active	Nov 12, 2023 - Nov 3, 2023	
 Charlie Davis	charledavis@gmail.com	873-844-9576	Stark Industries	#4b5563	Nov 12, 2023 - Nov 3, 2023	
 Charlie Baves	charlie.davis@email.com	873-853-8466	Stark Industries	#16a34a	Nov 12, 2023 - Nov 3, 2023	
 Eoron Ross	bobirbonen@gmail.com	874-883-2621	Acme Corp	#4b5563	Nov 12, 2023 - Nov 3, 2023	
 Johnn Ross	alicextlob@email.com	879-833-9228	Globex	#d97706	Nov 12, 2023 - Nov 3, 2023	
 Alice Green	bobd@eme@email.com	973-507-7886	Acme Corp	#16a34a	Nov 12, 2023 - Nov 3, 2023	
 Bob Ross	charlie.eavs@gmail.com	874-656-8831	Stark Industries	#d97706	Nov 12, 2023 - Nov 3, 2023	
 Bofo Ross	allicendavis@gmail.com	873-632-2837	Stark Industries	#d97706	Nov 12, 2023 - Nov 3, 2023	
 Charlie Davis	charliedastes@email.com	878-657-2566	Acme Corp	#16a34a	Nov 12, 2023 - Nov 3, 2023	
 Chaile Davis	infor@.detes@email.com	874-657-9921	Acme Corp	#d97706	Nov 12, 2023 - Nov 3, 2023	

Showing 1 to 10 of 150 entries

Previous

1

2

3

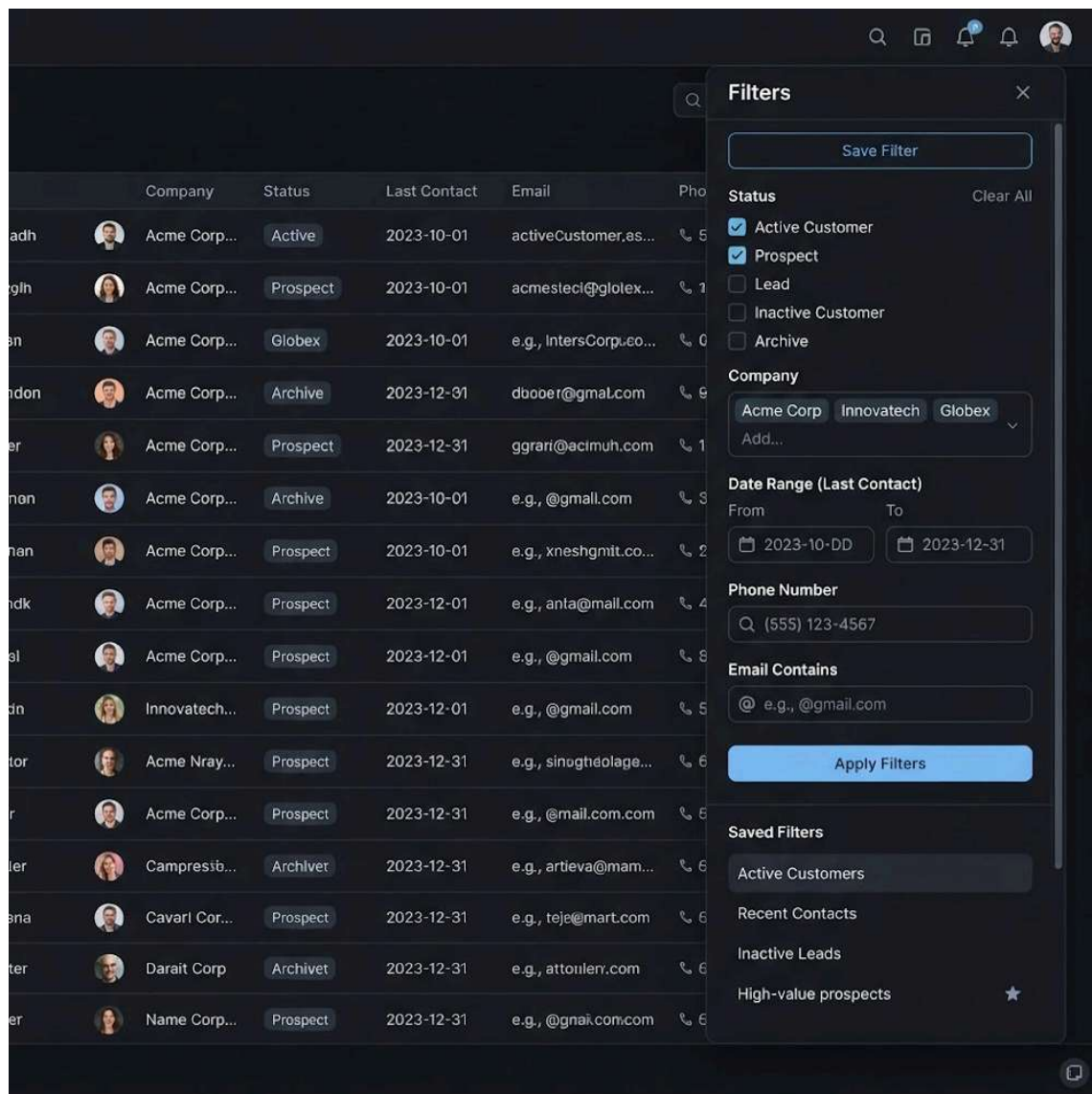
...

15

Next

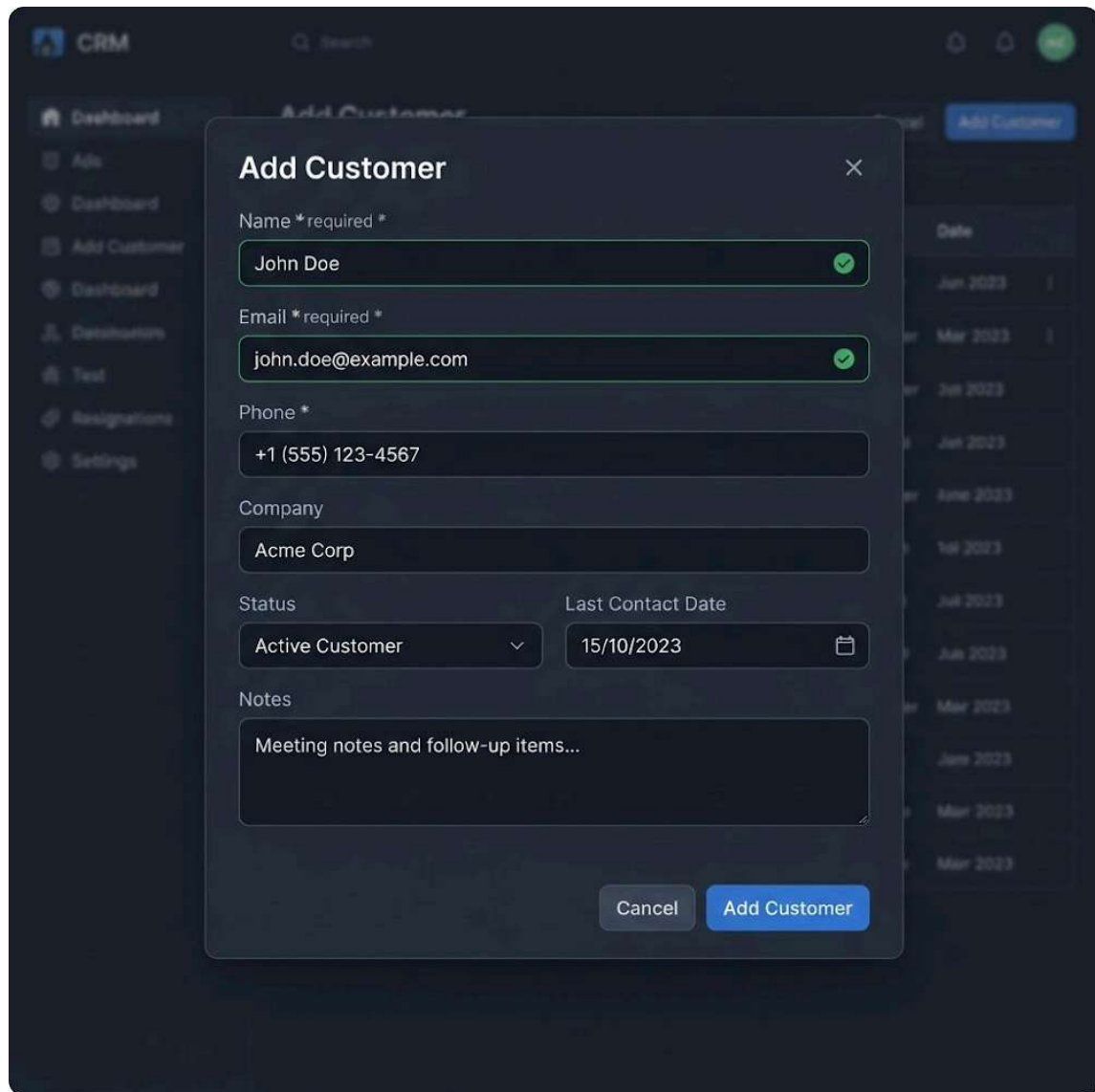
2. Advanced Filters Panel ★ (Key Feature)

- Sidebar or modal filter interface with multiple filter types:
 - **Status Filter** — Checkboxes (Active, Inactive)
 - **Company Filter** — Multi-select dropdown
 - **Date Range Filter** — Last contact date range picker
 - **Phone Number Filter** — Search/partial match
 - **Email Filter** — Search/partial match
- **Save Custom Filters** — Users can save filter combinations with names
- **Pre-built Filter Templates** — Examples: "Active Customers", "Recent Contacts", "Inactive Leads"
- **Clear All Filters** button
- **Apply Filters** button (or real-time as user selects)
- Filters persist and work seamlessly with search
- Visual indicator showing active filters count



3. Customer Details & Management

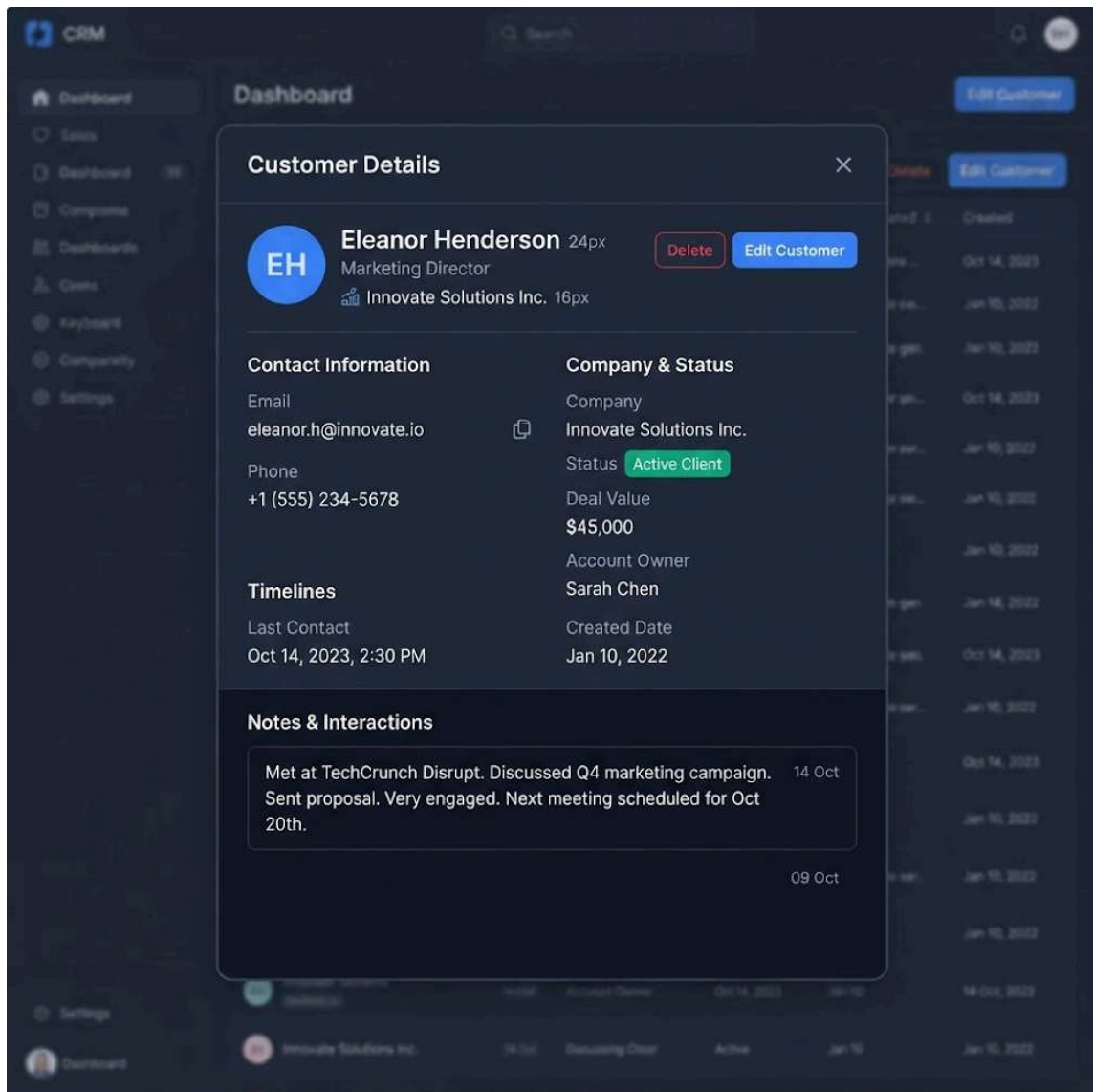
- Click a customer to view full details in modal/drawer
- Edit customer information with form validation
- Add new customer with validation
- Delete customer with confirmation dialog
- Update last contact date
- Customer notes field



The image shows a CRM application interface with a sidebar on the left containing menu items: Dashboard, Add, Add Customer, Dashboard, Dashboard, Dashboard, Test, Resignations, and Settings. The main area displays a modal titled "Add Customer" with a close button (X) in the top right corner. The form fields are as follows:

- Name *required ***: Input field with "John Doe" and a green checkmark.
- Email *required ***: Input field with "john.doe@example.com" and a green checkmark.
- Phone ***: Input field with "+1 (555) 123-4567".
- Company**: Input field with "Acme Corp".
- Status**: Dropdown menu with "Active Customer" selected.
- Last Contact Date**: Date picker showing "15/10/2023".
- Notes**: Text area with the placeholder "Meeting notes and follow-up items...".

At the bottom of the modal are two buttons: "Cancel" and "Add Customer".



4. Data Fetching with TanStack Query

- Use React Query for all data operations
- Implement proper caching with stale time
- Show loading/error states
- Refetch after mutations (add/edit/delete)
- Use hooks: `useQuery`, `useMutation`, `useQueryClient`
- Optimistic updates (optional bonus)

5. Form Validation & User Feedback

- Use shadcn/ui Form components for all forms

- Validate: email format, phone format, required fields
- Show validation errors inline
- Success/error toast notifications
- Disable buttons during submission

6. Drag & Drop Feature

- Implement drag-and-drop for reordering customers in the list, OR
- Drag-and-drop to move customers between custom filter groups, OR
- Drag-and-drop in the filters panel (reorder saved filters)
- Must use a proper drag-and-drop library (not HTML5 basic drag)

Nice-to-Have Bonus Features

- Bulk actions (select multiple, bulk edit status, bulk delete)
- Export filtered customers as CSV
- Dark/light mode toggle
- Keyboard shortcuts (e.g., Cmd+K to open filters)
- Debounced search for performance

Notes for Candidates

- Focus on code quality and architecture over fancy UI
- Commit regularly with meaningful commit messages
- Document any assumptions or design decisions
- If you get stuck, move on and come back (don't spend 1 hour on one feature)
- Test your filters thoroughly — this is critical
- Write comments for complex logic

Good luck! We're excited to see what you build! 🚀